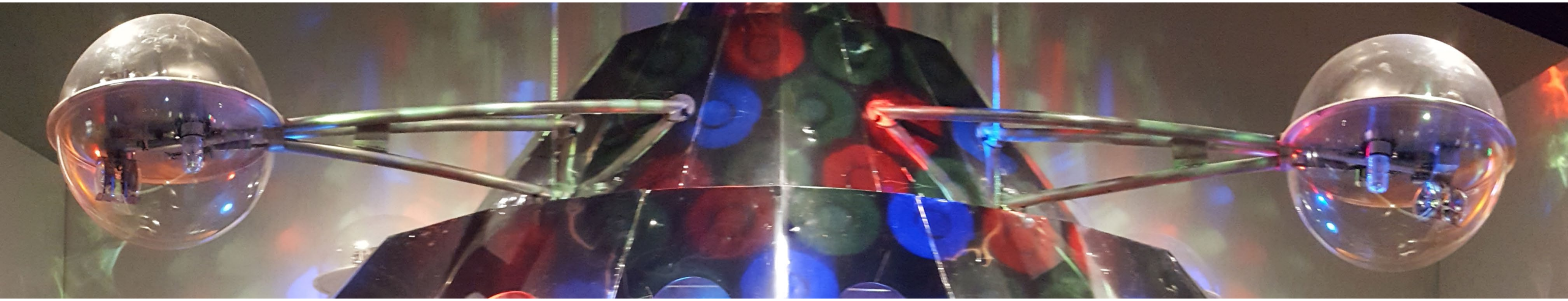
MINNESOTA IT SERVICES
DEPARTMENT OF NATURAL RESOURCES



We Want the Func!

Jeff Reinhart | GIS Programmer Analyst

Photo credit: Wikimedia Commons

What is an Azure Function App?

- Serverless
- Event driven
  - HTTP requests
  - Timers
  - Various other events
- Multiple languages
  - C#
  - JavaScript
  - PowerShell
  - Java
  - Python
  - TypeScript
  - Go













Photo credit: Wikimedia Commons

Why use an Azure Function App?

- Azure native
- No server to manage
- Low cost
- Elastic scaling
- Well documented



What do we use Function Apps for?

- Python API between web apps and data sources
  - Acts as a REST endpoint – similar to an ArcGIS Server service
  - Call routes via HTTP requests
  - Pass parameters to routes as needed
  - For example – Watershed Health Assessment Framework
- Authentication for Extranet apps
- Aggregator apps – for example – DNR Info tool:
  - Calls multiple secured ArcGIS services (FastAPI & asyncio)
  - Aggregates responses for frontend consumption and control



Photo credit: Wikimedia Commons

```python
@app.get('/location/{x}/{y}', tags=[Tags.general_purpose])
async def location(
    x: Annotated[float, Path(description=config_api.descCoordX)],
    y: Annotated[float, Path(description=config_api.descCoordY)]
):
    """
    Return location information for a given (x, y) coordinate pair.\n
    JSON object returned containing watershed and county names, IDs and area.\n
    Example: <a href="location/-10362030/5614607" target="_blank">location/-10362030/5614607</a>
    """
    def queryGeom(class_obj, x, y):
        """
        Query intersecting polygons for coordinates.
        """
        data = session.query(
            class_obj
        ).filter(
            class_obj.shape.ST_Intersects(
                sqlalchemy.func.ST_GeomFromText(
                    'Point({} {})'.format(x, y),
                    3857
                )
            )
        ).first()
        # if no result return object with None, else return ID.
        if data is None:
            out=dict(
                id=None
            )
        else:
            out = data.info()
        return out
    try:
        # START SESSION
        session = conn.startSessionWithSchema(dbSpatialSessMaker)
        # QUERY
        out = {}
        out['basin'] = queryGeom(classes.Basins, x, y)['id']
        out['catchment'] = queryGeom(classes.Catchments, x, y)['id']
        out['county'] = queryGeom(classes.Counties, x, y)['id']
        out['major'] = queryGeom(classes.Majors, x, y)['id']
        out['query_coordinates'] = [str(x), str(y)]
        out['upstream'] = out['catchment']
        # CLOSE SESSION AND RETURN
        session.close()
        return out
    except Exception as e:
        session.close()
        raise HTTPException(status_code=500, detail=f'Internal server error: {str(e)}')
```

Why? Customized GIS

- Enable POSTGIS on PG Flex Server
- For Watershed Health Assessment Framework: Explorer map click
- Intersects a point with five different datasets and returns attributes
- What would have been 5 requests to ArcGIS Server services with a complex JSON return is now one request with a simplified return


```
{
  "basin": "0401",
  "catchment": "0300100",
  "county": "27137",
  "major": 3,
  "query_coordinates": [
    "-10257588.0",
    "5905415.0"
  ],
  "upstream": "0300100"
}
```

How? Learning

- https://learn.microsoft.com/
- Overall understanding
  - Search for “azure function python http trigger”
  - Last checked – article was named “Azure Functions HTTP trigger”
  - Make sure language is set to Python
- Using VS Code extension:
  - Search for “quickstart create azure function vs code python”
  - “Quickstart: Create and deploy function code to Azure using Visual Studio Code”

How? VS Code

- Install:
  - Azure CLI
  - Azure Tools extension
  - Python version that will match your Azure Resource
- Start local project
  - Command Palette – Azure Functions: Create New Project (wizard)
  - Run project locally
  - Stop! Deploy the template first.



Photo credit: Wikimedia Commons

Azurite: Start F5 (run)

How? VS Code

function_app.py

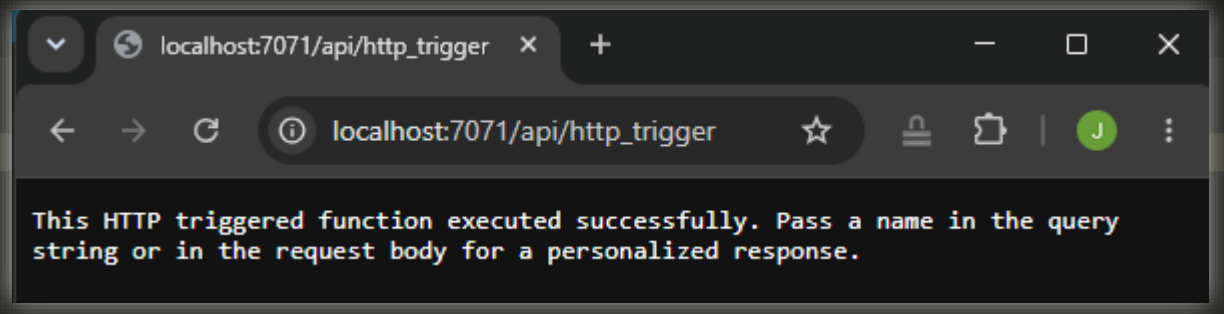
```python
import azure.functions as func
import logging

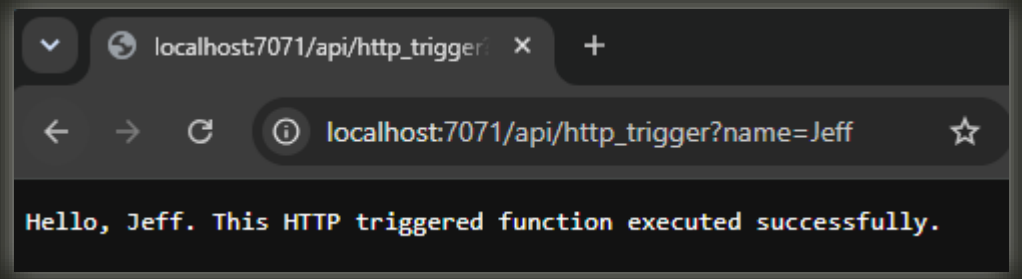
app = func.FunctionApp(http_auth_level=func.AuthLevel.ANONYMOUS)

@app.route(route="http_trigger")
def http_trigger(req: func.HttpRequest) -> func.HttpResponse:
    logging.info('Python HTTP trigger function processed a request.')

    name = req.params.get('name')
    if not name:
        try:
            req_body = req.get_json()
        except ValueError:
            pass
        else:
            name = req_body.get('name')

    if name:
        return func.HttpResponse(f"Hello, {name}. This HTTP triggered function executed successfully.")
    else:
        return func.HttpResponse(
            "This HTTP triggered function executed successfully. Pass a name in the query string or in the request body for a personalized response.",
            status_code=200
        )
```





How? Azure Resources

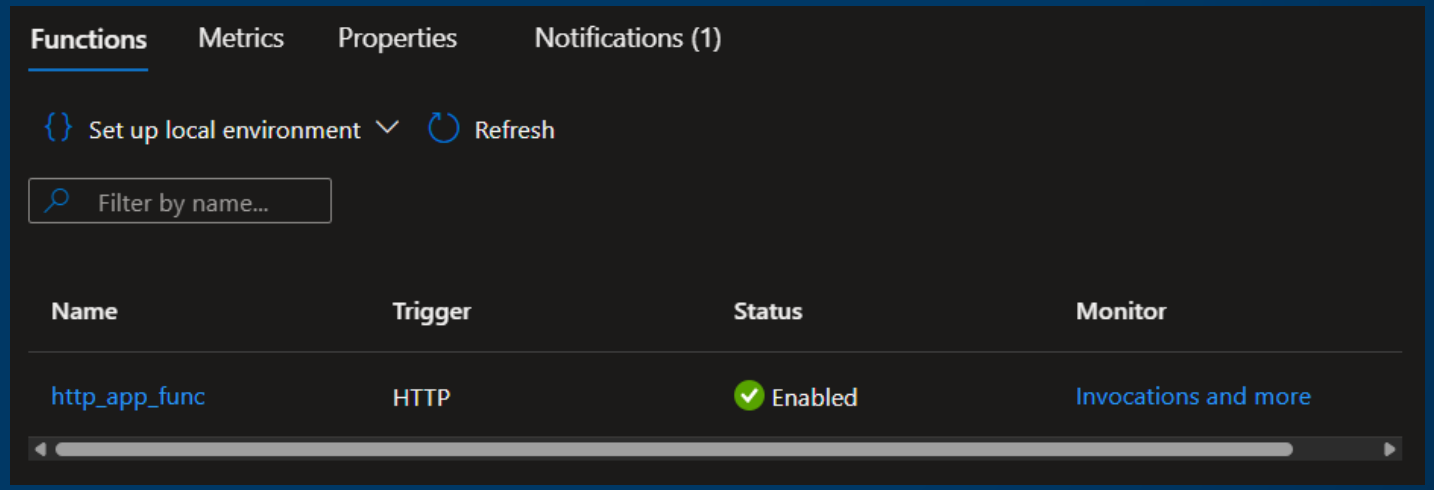
- Deploy resources
  - Azure Portal (web GUI)
  - Bicep
  - Azure Resource Manager Templates
- Function App includes:
  - Function App (can have multiple slots)
  - Hosting Plan
  - Storage Account
  - Log Analytics Workspace / Application Insights (optional)
  - Private Endpoints (if on a virtual network)

```
resource functionApp 'Microsoft.Web/sites@2024-04-01' = {
  name: functionAppName
  location: location
  kind: 'functionapp,linux'
  tags: tags
  identity: {
    type: 'SystemAssigned'
  }
  properties: {
    reserved: true
    serverFarmId: hostingPlanId
    siteConfig: {
      linuxFxVersion: linuxFxVersion
      appSettings: [
        // {
        //   name: 'APPINSIGHTS_INSTRUMENTATIONKEY'
        //   value: applicationInsights.properties.InstrumentationKey
        // }
        {
          name: 'AzureWebJobsStorage'
          value: 'DefaultEndpointsProtocol=https;AccountName=${storageAccountName};EndpointSu
        }
        {
          name: 'WEBSITE_CONTENTAZUREFILECONNECTIONSTRING'
          value: 'DefaultEndpointsProtocol=https;AccountName=${storageAccountName};EndpointSu
        }
        {
          name: 'WEBSITE_CONTENTSHARE'
          value: '${toLower(functionAppName)}-p1' // since these will swap back and forth bet
        }
        {
          name: 'FUNCTIONS_EXTENSION_VERSION'
          value: '~4'
        }
        {
          name: 'FUNCTIONS_WORKER_RUNTIME'
          value: functionWorkerRuntime
        }
      ]
    }
    httpsOnly: true
  }
}
```

How? Deploy

- Methods:
  - VS Code - Azure Tools UI
  - Azure CLI
- What happens in Azure?
  - Installs requirements.txt
  - Builds your code in the Storage Account
  - Queries your routes

- Deploy the template first
  - If not, app may never initialize properly
  - Needs to do a successful route query
  - Important if adding FastAPI or Flask
  - In Azure Portal | Function App | Overview:



How? Local Dev & Testing

- Add libraries to requirements.txt
- Azurite storage emulator
  - Azurite: Start
  - Locally emulates Azure Storage Account that is deployed with your Function App
- Run your code! (F5)
  - Updates virtual environment
  - Refresh is not great – has to restart each time
  - Logs will tell you where to go for localhost and port



Photo credit: Wikimedia Commons

```python
@app.get('/location/{x}/{y}', tags=[Tags.general_purpose])
async def location(
    x: Annotated[float, Path(description=config_api.descCoordX)],
    y: Annotated[float, Path(description=config_api.descCoordY)]
):
    """
    Return location information for a given (x, y) coordinate pair.\n
    JSON object returned containing watershed and county names, IDs and area.\n
    Example: <a href="location/-10362030/5614607" target="_blank">location/-10362030/5614607</a>
    """
    def queryGeom(class_obj, x, y):
        """
        Query intersecting polygons for coordinates.
        """
        data = session.query(
            class_obj
        ).filter(
            class_obj.shape.ST_Intersects(
                sqlalchemy.func.ST_GeomFromText(
                    'Point({} {})'.format(x, y),
                    3857
                )
            )
        ).first()
        # if no result return object with None, else return ID.
        if data is None:
            out=dict(
                id=None
            )
        else:
            out = data.info()
        return out
    try:
        # START SESSION
        session = conn.startSessionWithSchema(dbSpatialSessMaker)
        # QUERY
        out = {}
        out['basin'] = queryGeom(classes.Basins, x, y)['id']
        out['catchment'] = queryGeom(classes.Catchments, x, y)['id']
        out['county'] = queryGeom(classes.Counties, x, y)['id']
        out['major'] = queryGeom(classes.Majors, x, y)['id']
        out['query_coordinates'] = [str(x), str(y)]
        out['upstream'] = out['catchment']
        # CLOSE SESSION AND RETURN
        session.close()
        return out
    except Exception as e:
        session.close()
        raise HTTPException(status_code=500, detail=f'Internal server error: {str(e)}')
```

How? Routes

- Decorator: @app.get()
  - Route path, data validation, doc config, security
- Route function: location()
  - Call parameters, headers, doc config
- Return dict
  - Gets serialized as JSON
- Error handling
  - Close sessions, cleanup, fail meaningfully

How? FastAPI

- For larger APIs, recommend using FastAPI
- Uses asyncio
- OOTB Swagger Docs
- Data Validation
- Search “use fastapi in azure function app”

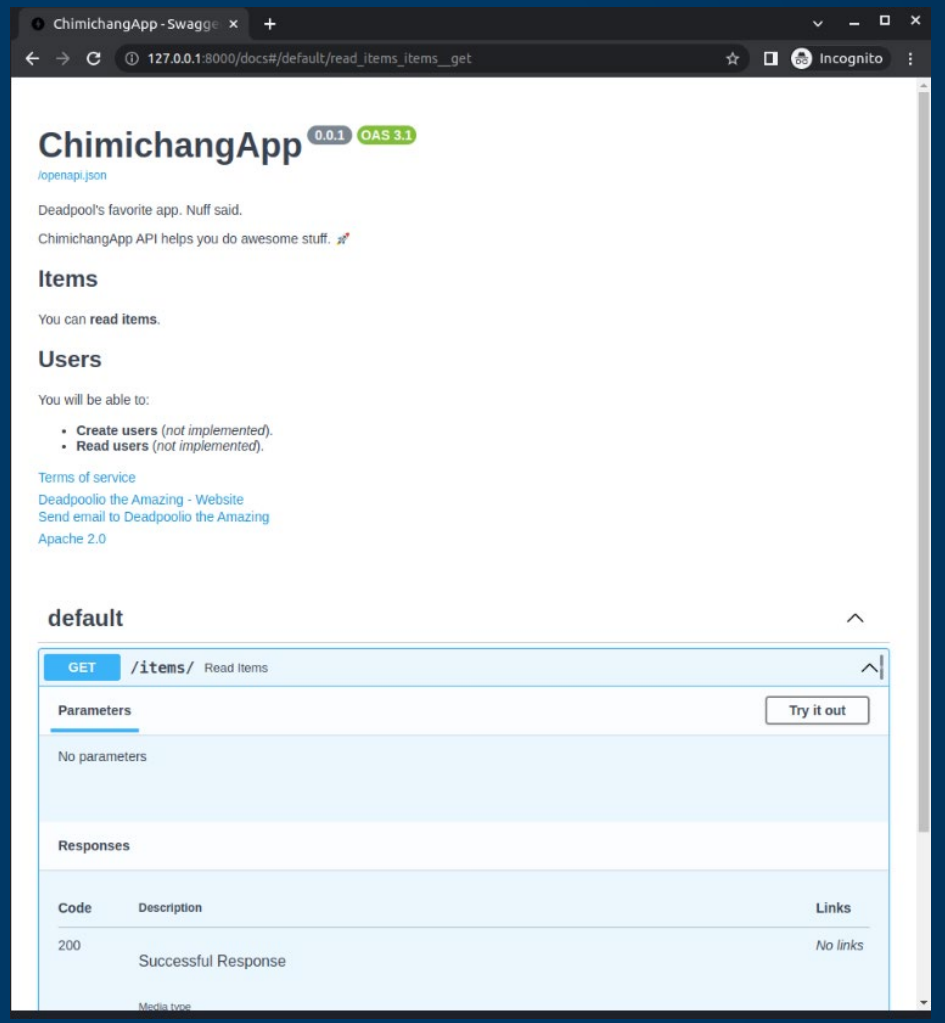


Photo credit: https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-api

Issues – Deployment Black Box

- Deploy runs code to query HTTP triggers (routes)
- If anything errors when deploying – there are NO LOGS with the error
- Error Indicators:
  - In Azure Portal – no routes in Azure Portal | Function App | Overview
  - Route will 404

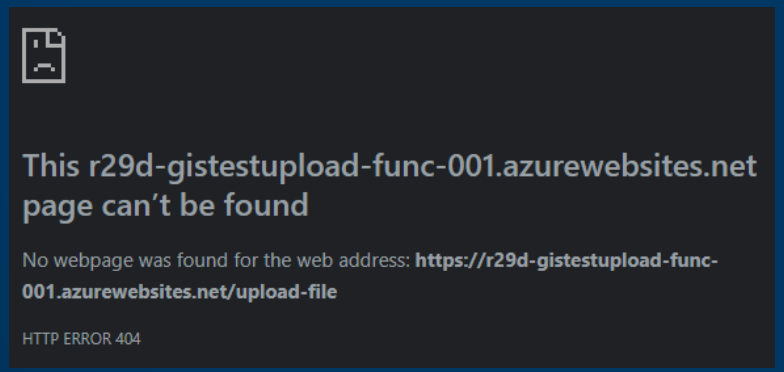




Photo credit: Wikimedia Commons

Issues – Deployment Black Box

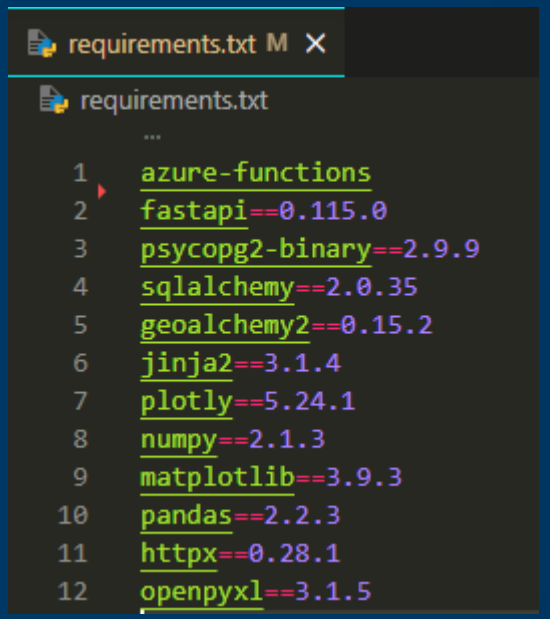
- Fixes:
  - Prune code and deploy again until it works
  - Move potentially offending code inside a route
- Upvote a permanent fix!



Photo credit: Wikimedia Commons

Issues – Environment

- No access to operating system
  - No arcpy
  - psycopg2-binary OR psycopg (3) OR asyncpg
  - Watch library dependencies
- OS is Linux – directory with no files does not exist
- Naming resources
  - Slot names cannot be over 32 characters
  - Naming collision in the storage account



Issues – Environment

- Virtual Network
  - State of MN Policy (best practice): Postgres Flex Servers on VNET
  - To connect, Function App also on VNET
  - Private Endpoints for Storage Account blob, file, table resources
- Firewall Rules… allow:
  - pypi.python.org
  - pypi.org
  - pythonhosted.org
  - files.pythonhosted.org
  - oryx-cdn.microsoft.io



Photo credit: Wikimedia Commons

MINNESOTA IT SERVICES

Thank You!

Jeff Reinhart

jeff.reinhart@state.mn.us